Multiscale Alignment and Progressive Feature Fusion Network for High Resolution Remote Sensing Images Change Detection

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***Abstract*— The advancement of deep learning technology has significantly improved the performance of high-resolution remote sensing (HRRS) image change detection (CD) task. HRRS images on CD are usually captured under varying conditions, which may generate pseudo-changes caused by seasonal variations, changes in lighting angles, and object motion. These factors can interfere with the effective modeling of difference information. However, conventional methods of modeling difference information may not adequately address these challenges, leaving them susceptible to such factors. Therefore, this article proposes a multiscale alignment and progressive feature fusion network (MAPNet) based on convolutional neural network (CNN) and transformer, which can effectively model difference information. First, a flow- and attention-guided bitemporal alignment module (FA-BAFM) is designed to align bitemporal features and capture the differences between them. Second, a progressive difference feature fusion module (PDFFM) is developed to comprehensively fuse the difference information across various scales and levels. Finally, a local feature enhancement module (LFEM) is constructed to improve the ability of the backbone to extract both global and local features. Experimental results on the learning, vision, and remote sensing CD (LEVIR-CD), Wuhan University (WHU), and Sun Yat-Sen University CD (SYSU-CD) datasets show that MAPNet achieves F1-scores of 91.90%, 94.19%, and 83.69%, respectively, outperforming the state-of-the-art (SOTA) methods. The demo code will be released at https://github.com/zlin9/mapnet.**

***Index Terms*— Change detection (CD), convolutional neural network (CNN), feature alignment, remote sensing (RS), transformer.**

I. Introduction

CHANGE detection (CD), as one of the important research topics in the remote sensing (RS) community, aims to identify changed areas by analyzing two (or more) RS images covering the same location and collected at different times [1], [2]. In recent years, with the development of sensors and technology, the resolution of satellite RS images has constantly improved, garnering increased attention for high-resolution RS (HRRS) images in the field of CD. This is attributed to the wealth of change information that can be derived from HRRS images, making it valuable for numerous practical applications, such as damage assessment [3], urban studies [4], ecosystem monitoring [5], agricultural surveying [6], and resource management [7].

The continuous advancements in deep learning techniques and RS images have propelled methodologies based on deep learning to the forefront as the primary solutions for challenges in CD [8]. In accordance with the point of view presented in research [9], deep learning-based CD frameworks can be categorized into two types: single-stream and double-stream. In the single-stream framework, an early-fusion strategy is usually used to combine the bitemporal image pair before proceeding with any further processing steps. This allows the image segmentation algorithms to be easily adapted to address the CD problem with minor modifications [10]. In the double-stream framework, the Siamese network emerges as a noteworthy architecture. It is distinguished by its unique dual-input structure, which accommodates the flexibility of employing shared or nonshared weights. This versatility enables the effective handling of bitemporal images. In [11], the fully convolutional Siamese networks were first applied in CD for RS images. Similarly, other enhanced algorithms utilizing Siamese networks, such as [12], [13], and [14], differ in their approach to feature processing. In [12], a hierarchically structured transformer encoder and a multilayer perceptron decoder were combined in a Siamese network architecture to effectively capture the multiscale long-range details necessary for accurate CD. In [13], spatial–temporal attention modules were developed to capture long-range spatial–temporal dependencies to improve feature representations. Feng et al. [14] integrated self-attention and cross-attention to steer the global feature distribution of each input, promoting information coupling between interlevel representations and suppressing task-irrelevant interferences.

Although existing CD methods have shown advantages in various scenarios, there are still limitations in modeling difference information. The methods for modeling the differences between input pairs typically include either direct methods (such as difference operations and channel concatenation) or the implicit embedding of feature interactions. While these methods are simple and efficient, they overlook the impact of pseudo-changes when capturing difference information.

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Additionally, due to the complex texture information and limited spectral information of HRRS images, dealing with complex change scenarios becomes challenging [15]. Therefore, it is crucial to mitigate the influence of pseudo-changes and ensure accurate CD results.

In view of this, an innovative network designed for CD is proposed, named the multiscale alignment and progressive feature fusion network (MAPNet). First, a difference capture module named flow and attention-guided bitemporal alignment fusion module (FA-BAFM) is designed to mitigate the impact of pseudo-changes by feature alignment. Second, after obtaining the difference information, we perform progressive feature fusion on multiple scales and across levels using the progressive difference feature fusion module (PDFFM), which improves adaptability in complex change scenarios. Finally, we adopt a convolutional neural network (CNN)-transformer combination variant to extract rich global and local features. Notably, the multiscale structure of the network not only enriches the diversity of difference information but also establishes a more effective framework for deep supervision at different levels. The primary contributions of this article can be summarized as follows.

1) A plug-and-play FA-BAFM is proposed for acquiring multiscale difference information. Specifically, aligning bitemporal features can effectively mitigate interference caused by pseudo-changes and improve precision in extracting difference information.
2) The PDFFM consists of the progressive atrous fusion module (PAFM) and the interlevel alignment fusion module (IAFM). It utilizes a coarse-to-fine connection to integrate difference information from various scales and levels and enhance the ability to deal with complex change scenarios.
3) The evaluation results from three publicly available CD datasets [Sun Yat-Sen University CD (SYSU-CD), learning, vision, and remote sensing CD (LEVIR-CD), and Wuhan University (WHU)] demonstrate that the proposed MAPNet achieves superior qualitative and quantitative outcomes.

The remaining sections of this article are organized as follows. Section II provides a review of the relevant literature on CD. Section III provides a detailed description of the proposed method. Section IV outlines the experimental setup and presents both qualitative and quantitative results. Finally, the conclusions are presented in Section V.

II. Related Works

A. CNN-Based CD Methods

The CD task can essentially be viewed as a pixel classification task like image segmentation. Encoder–decoder architecture networks, such as SegNet [16] and U-Net [17], have been proven to be suitable for image segmentation tasks as well. Based on this, Daudtt et al. [11] introduced three full convolution (FC) encoder–decoder architectural networks: FC-early fusion (FC-EF), FC-Siam-Conc, and FC-Siam-Diff. FC-EF utilizes an early fusion approach that stacks biphasic images as inputs to the network. Both FC-Siam-Conc and FC-Siam-Diff extract multiscale features using a Siamese network and then connect the cascaded features and feature differences to the decoder, respectively. Daudt et al. [18] replaced the convolutional layers in FC-EF with residual blocks to improve the accuracy of CD while optimizing network training. However, the long-distance connections in the U-Net overlook semantic differences between features, which affects the quality of the fused features. To reduce semantic differences between features, Zhou et al. [19] introduced UNet++. This network gradually diminishes semantic differences through dense hopping connections and has been successfully applied in CD task, such as Peng et al. [20] introduced the multiple side-output fusion strategy, and Fang et al. [21] proposed the densely connected Siamese network (SNUNet). Li et al. [22] utilized two parallel UNet++ to process high-resolution image features and joint features of multispectral and SAR images, respectively, for CD in multisource RS data. Additionally, Wu et al. [23] proposed a multitask CD framework with differential band reweighting and unbalanced contrastive learning, enhancing CD by extracting discriminative information and addressing sample imbalance using multiple prototypes. Luo et al. [24] proposed the multiscale difference variation feature fusion network (MSDFFN), which enhances feature representation by learning the fine variation components between bitemporal hyperspectral images at different scales. However, the methods relying on CNN have limitations due to the size of the receptive field and a lack of attention to global information.

B. Transformer-Based CD Methods

Transformer-based models demonstrate robust global feature extraction capabilities, usually achieving comparable or even superior performance compared with convolutional models. Rich global information enhances the ability of networks to correlate bitemporal features, aiding in the analysis and interpretation of changing patterns [25]. Chen et al. [26] were the first to integrate the transformer structure into CD, improving feature expression by using semantic tokens with global context information. Shi et al. [27] integrated information from different time points and spatial locations at the token level and modeled global dependencies in both spatial and temporal dimensions. However, these two methods exclusively capture the global dependency of single-scale features while overlooking the multiscale spatial information. Bandara and Patel [12] constructed a pure transformer Siamese network (ChangeFormer) to extract multiscale features. Zhang et al. [28] replaced the convolutional blocks with stacked Swin-transformer modules [29] in the encoder–decoder, forming a pure transformer network. To harness the benefits of both CNN and the transformer, Feng et al. [30] proposed an intrascale cross-interaction and interscale feature fusion network (ICIF-Net). This network extracts local and global features from the bitemporal images by integrating ResNet18 [31] and pyramid vision transformer v2 (PvTv2) [32] through a parallel structure. Zhang et al. [15] proposed an asymmetric cross-attention hierarchical network (ACAHNet), which also combines CNN and transformer in a series–parallel manner to enhance the interaction between features extracted from the CNN and the transformer.

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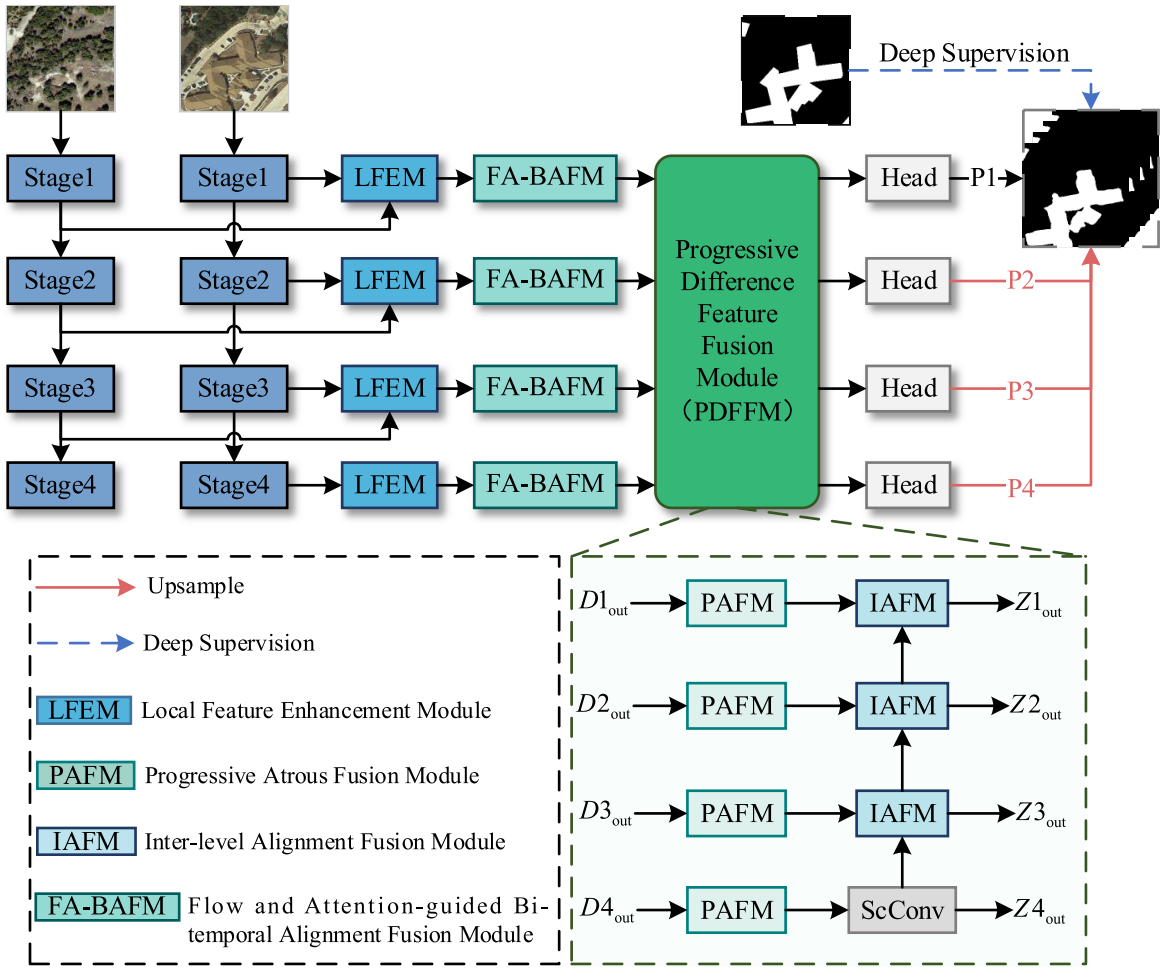


Fig. 1. Framework of the proposed MAPNet. Stage 1–Stage 4 represent four feature extraction stages of the MiT backbone. The green dotted box represents the detailed structure of the PDFFM. The head represents the prediction head. P1 represents supervision for MAPNet, and P2–P4 represent deep supervision for MAPNet.

III. Methodology

The overall architecture of the MAPNet, which includes the local feature enhancement module (LFEM), flow and attention-guided bitemporal alignment module (FA-BAFM), and PDFFM, is elucidated in Section III, as shown in Fig. 1. First, the bitemporal images are fed into the backbone network to extract bitemporal features across four stages with distinct scales. Second, these features are further enhanced through LFEM, which extracts local information. After obtaining the enhanced bitemporal features, the difference information between them could be captured by the FA-BAFM, which employs an effective difference information modeling approach. Then, the PDFFM fuses this difference information from multiple scales and across levels, helping the network adapt to complex change scenarios. Finally, the fused difference features are encoded by conventional prediction heads and then combined with the argmax operation to generate the prediction map. It should be noted that we apply a deep supervision approach to enhance the learning capability and convergence speed of the network. Output results from the second, third, and fourth stages serve as learning targets for deep supervision. Overall, the MAPNet emphasizes the acquisition of difference information and the integration of difference information across various scales and levels, both of which are critical for the CD task.

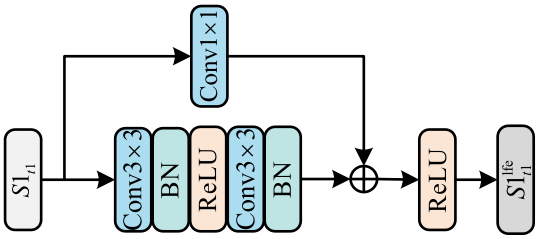


Fig. 2. Illustration of the proposed LFEM.

A. LFEM Module

To complement the global attention of the transformer with the powerful local sensing capability of CNN, the LFEM is proposed, primarily comprising convolutional (Conv) layers, batch normalization (BN) layers [33], and rectified linear unit (ReLU) activation function, as depicted in Fig. 2. The ReLU activation function adds nonlinearity, and the interleaved BN layers stabilize the training process. The LEFM begins with a sequence of Conv-BN-ReLU-Conv-BN, branches out with a single convolution, and concludes with a ReLU layer. Please note that despite the structural similarity to ResNet [31], the

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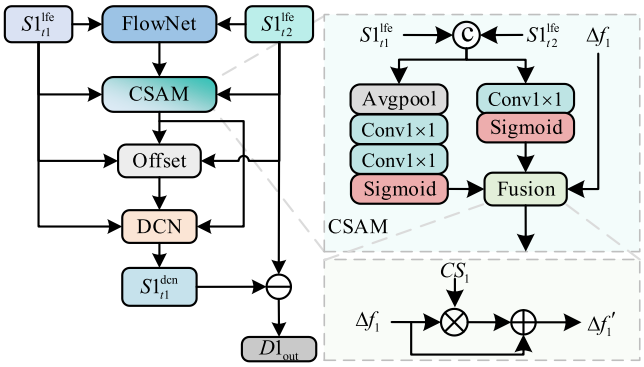


Fig. 3. Illustration of the proposed FA-BAFM.

design purposes in the LFEM are distinct. We utilize the residual connection to preserve sufficient spatial details rather than solely addressing degradation issues in deep networks. The initial bitemporal features are extracted at four stages as shown in Fig. 1, taking the first moment as an example, i.e., $S1_{t1}$, $S2_{t1}$, $S3_{t1}$, and $S4_{t1}$, and the mentioned features are used as inputs of the LFEM

$$\begin{aligned} S1_{t1}^{\text{lfe}} &= \text{LFEM}(S1_{t1}) \\ S2_{t1}^{\text{lfe}} &= \text{LFEM}(S2_{t1}) \\ S3_{t1}^{\text{lfe}} &= \text{LFEM}(S3_{t1}) \\ S4_{t1}^{\text{lfe}} &= \text{LFEM}(S4_{t1}) \end{aligned} \tag{1}$$

where $S1_{t1}^{\text{lfe}}$, $S2_{t1}^{\text{lfe}}$, $S3_{t1}^{\text{lfe}}$, and $S4_{t1}^{\text{lfe}}$ denote the features through the LFEM at the four stages. Specifically, when the $S1_{\text{t1}}$ is fed into the LFEM, the process can be formulated as

$$S1_{t1}^{\text{lfe}} = \text{ReLU}\left(S1_{t1}^{\text{temp}} + \text{Conv}_{1\times1}(S1_{t1})\right) \tag{2}$$

$$S1_{t1}^{\text{temp}} = \text{BN}(\text{Conv}_{3\times3}(\text{ReLU}(\text{BN}(\text{Conv}_{3\times3}(S1_{t1}))))) \tag{3}$$

where $S1_{t1}^{\text{temp}}$ is an intermediate variable. $\text{Conv}_{3\times3}(\cdot)$ and $\text{Conv}_{1\times1}(\cdot)$ represent the convolution operations with 3×3 kernel and 1×1 kernel, respectively. $\text{BN}(\cdot)$ denotes the BN layer and $\text{ReLU}(\cdot)$ denotes the ReLU activation function.

B. FA-BAFM Module

To alleviate the disturbance of pseudo-changes and capture the difference information effectively, the FA-BAFM is proposed, primarily consisting of a deformable convolution (DCN) [34] guided by optical flow and attention mechanism and an offset layer [35], as illustrated in Fig. 3. Specifically, the bitemporal feature maps are resampled through the DCN to compensate for the nonsemantic difference [36]. Then, the difference information between the bitemporal feature maps is captured by computing their subtraction. First, the optical flow between the bitemporal features $S1_{t1}^{\text{lfe}}$ and $S1_{t2}^{\text{lfe}}$ is captured by the FlowNet (2)-layer Conv). This process is formulated as

$$\Delta f_1 = \text{FlowNet}\left(S1_{t1}^{\text{lfe}}, S1_{t2}^{\text{lfe}}\right) \tag{4}$$

where Δf_1 represents the optical flow between bitemporal features, which can characterize the pixel displacement between features at different time instants [37].

To guide the optical flow focusing on the changing regions, the channel and spatial attention module (CSAM) is designed, which consists of a channel attention module (CAM) and a spatial attention module (SAM), as shown in Fig. 3. The CAM consists of an avg-pooling layer, two 1×1 convolution layers, and a Sigmoid layer. The SAM involves a 1×1 convolution layer and a Sigmoid layer. These two modules focus on learning channel-wise and spatial-wise attention distribution vectors of difference information, respectively. The $S1_{t1}^{\text{lfe}}$ and $S1_{t2}^{\text{lfe}}$ are the inputs of CSAM, and CS_1 is the target attention vectors. The process of the CSAM is formulated as

$$CS_1 = C_1 \otimes S_1 \tag{5}$$

$$C_1 = \sigma\left(\text{Conv}_{1\times1}\left(\text{Conv}_{1\times1}\left(\text{Avg}\left(\text{ConCat}\left(S1_{t1}^{\text{lfe}}, S1_{t2}^{\text{lfe}}\right)\right)\right)\right)\right) \tag{6}$$

$$S_1 = \sigma\left(\text{Conv}_{1\times1}\left(\text{ConCat}\left(S1_{t1}^{\text{lfe}}, S1_{t2}^{\text{lfe}_j}\right)\right)\right) \tag{7}$$

where $\otimes$ denotes the element-wise multiplication, C_1 represents the channel attention features, and S_1 represents the spatial attention features. $\sigma(\cdot)$ is the Sigmoid function, and $\text{Avg}(\cdot)$ denotes the operation of avg-pooling. $\text{ConCat}(\cdot)$ denotes the operation of concatenation. Then, we can capture the feature $\Delta f_1'$ through the CSAM. The process is formulated as

$$\Delta f_1' = \Delta f_1 + (\Delta f_1 \otimes CS_1)) \tag{8}$$

where $+$ represents the element-wise sum operator. Second, the offsets can be obtained using the following expression:

$$\Delta O = \text{Offset}\left(S1_{t1}^{\text{lfe}}, S1_{t2}^{\text{lfe}}, \Delta f_1'\right) \tag{9}$$

where $\text{Offset}(\cdot)$ represents the offset layer, and ΔO represents the offsets. The alignment operation can be achieved by DCN. This procedure can be formulated as

$$S1_{t1}^{\text{dcn}} = \text{DCN}\left(S1_{t1}^{\text{lfe}}, \Delta O, \Delta f_1'\right) \tag{10}$$

where $\text{DCN}(\cdot)$ denotes the DCN, and $S1_{t1}^{\text{dcn}}$ represents the aligned features. Finally, the difference feature can be obtained using the following process:

$$D1_{\text{out}} = \left|S1_{t2}^{\text{lfe}} - S1_{t1}^{\text{dcn}}\right| \tag{11}$$

where $D1_{\text{out}}$ represents the final difference feature of stage one. The $|\cdot|$ denotes the absolute operation. Similarly, through the above process, we can obtain the difference features of the other three stages, i.e., $D2_{\text{out}}$, $D3_{\text{out}}$, and $D4_{\text{out}}$, as shown in Fig. 1.

C. PDFFM Module

Feature fusion is an effective approach for integrating information across different scales and levels [38], [39]. In this work, we propose a PDFFM, mainly comprising two main components: PAFM and IAFM, as shown in Fig. 1. In this section, we provide a detailed description of the internal structure of the designed PDFFM.

1) PAFM: To integrate the difference information with various scales, the PAFM is designed, consisting of a max-pooling layer, two convolutional layers, and a sequence of atrous convolutions with various atrous rates, as illustrated in Fig. 4. The design of the PAFM is inspired by [40]. First, the significant difference information is stressed by the

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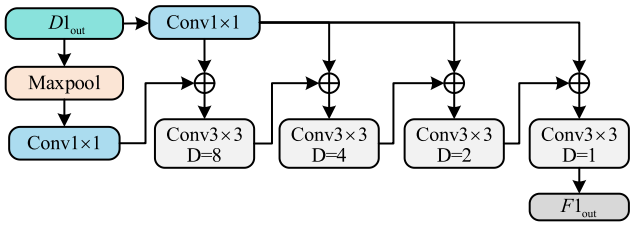


Fig. 4. Illustration of the proposed PAFM.

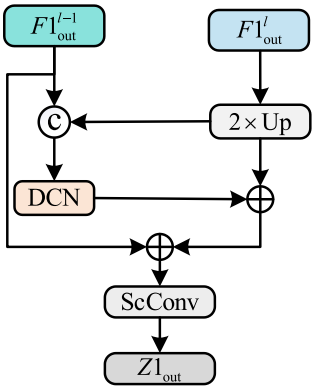


Fig. 5. Illustration of the proposed IAFM.

max-pooling layer and 1 × 1 convolution. Second, these features are element-wise summed with the original difference features and passed through a series of atrous convolutions. Notably, the atrous rate configuration of {8, 4, 2, 1} is applied to optimize the utilization of input features and achieve improved results [41]. The process of the PAFM can be formulated as

$$\begin{aligned} D1^{\mathrm{m}}_{\mathrm{out}} &= \mathrm{Conv}_{1\times1}(\mathrm{Max}(D1_{\mathrm{out}})) \\ D1^{d8}_{\mathrm{out}} &= \mathrm{Conv}_{1\times1}(D1_{\mathrm{out}}) + \mathrm{Conv}^{d8}_{3\times3}\left(D1^{\mathrm{m}}_{\mathrm{out}}\right) \\ D1^{d4}_{\mathrm{out}} &= \mathrm{Conv}_{1\times1}(D1_{\mathrm{out}}) + \mathrm{Conv}^{d4}_{3\times3}\left(D1^{d8}_{\mathrm{out}}\right) \\ D1^{d2}_{\mathrm{out}} &= \mathrm{Conv}_{1\times1}(D1_{\mathrm{out}}) + \mathrm{Conv}^{d2}_{3\times3}\left(D1^{d4}_{\mathrm{out}}\right) \\ F1_{\mathrm{out}} &= \mathrm{Conv}_{1\times1}(D1_{\mathrm{out}}) + \mathrm{Conv}^{d1}_{3\times3}\left(D1^{d2}_{\mathrm{out}}\right) \end{aligned} \tag{12}$$

where $D1_{\mathrm{out}}$ represents the difference features fed to the PAFM, and Max(·) denotes the operation of max-pooling. $\mathrm{Conv}_{1\times1}(\cdot)$ is the 1 × 1 convolution, and $\mathrm{Conv}^{d8}_{3\times3}(\cdot)$ denotes atrous convolution with an atrous rate of 8 with a 3 × 3 kernel. Similarly, $\mathrm{Conv}^{d4}_{3\times3}(\cdot)$, $\mathrm{Conv}^{d2}_{3\times3}(\cdot)$, and $\mathrm{Conv}^{d1}_{3\times3}(\cdot)$ represent the atrous convolutions with atrous rates of 4, 2, and 1 using a convolution kernel of 3 × 3, respectively. + denotes the element-wise sum, and $D1^{\mathrm{m}}_{\mathrm{out}}$ represents the difference feature after the max-pooling and 1×1 convolution. Finally, $D1^{d8}_{\mathrm{out}}$, $D1^{d4}_{\mathrm{out}}$, $D1^{d2}_{\mathrm{out}}$, and $F1_{\mathrm{out}}$ represent the output features under the atrous convolution corresponding to atrous rates {8, 4, 2, 1}, respectively. Similarly, through the above process, the features of the other three stages are obtained, i.e., $F2_{\mathrm{out}}$, $F3_{\mathrm{out}}$, and $F4_{\mathrm{out}}$.

2) IAFM: To alleviate the loss of boundary information during feature fusion from different levels, the IAFM is designed, which mainly consists of a DCN and a self-calibrated convolution (ScConv) [42], as illustrated in Fig. 5. The design of the IAFM is inspired by [43]. This module can learn transformation offsets of pixels to contextually align upsampled features, facilitating multilevel feature fusion effectively. $F1^{l-1}_{\mathrm{out}}$ and $F1^{l}_{\mathrm{out}}$ denote the inputs of the IAFM, where $F1^{l}_{\mathrm{out}}$ is from deeper levels. First, the upsampling operation is applied to address mismatches between $F1^{l-1}_{\mathrm{out}}$ and $F1^{l}_{\mathrm{out}}$. Second, we concatenate the upsampled $F1^{l}_{\mathrm{out}}$ and $F1^{l-1}_{\mathrm{out}}$ into the DCN to obtain the offset value of $F1^{l}_{\mathrm{out}}$, where each offset value can be viewed as the shifted distances in 2-D space [37] between each point in $F1^{l}_{\mathrm{out}}$ and its corresponding point in $F1^{l-1}_{\mathrm{out}}$. Finally, the features from different levels are fused by pixel addition. Finally, to reduce the feature redundancy and enhance feature representation capabilities during the fusion process, the ScConv is adopted in the IAFM. The process of IAFM can be formulated as

$$\begin{aligned} &Z1_{\mathrm{out}} \\ &= \mathrm{ScConv}\left(F1^{l-1}_{\mathrm{out}} + \mathrm{DCN}\left(\mathrm{ConCat}\left(F1^{l-1}_{\mathrm{out}}, F1^{l}_{\mathrm{out}}\right)\right) + F1^{l}_{\mathrm{out}}\right) \end{aligned} \tag{13}$$

where ScConv(·) is the ScConv layer, DCN(·) is the DCN, and ConCat(·) represents concatenate operation. $F1^{l-1}_{\mathrm{out}}$ and $F1^{l}_{\mathrm{out}}$ represent the inputs of IAFM from different levels, $Z1_{\mathrm{out}}$ denotes the output of IAFM. Similarly, after the above process, the output features from three stages are obtained, i.e., $Z2_{\mathrm{out}}$, $Z3_{\mathrm{out}}$, and $Z4_{\mathrm{out}}$, as shown in Fig. 1.

Here, we describe the advantages and differences between the proposed FA-BAFM and IAFM. First, the FA-BAFM employs optical flow and attention-guided feature alignment to capture multiscale difference information in bitemporal features. Unlike conventional alignment methods, we incorporate an attention mechanism to aid the alignment method in recognizing change regions. Second, compared to the FA-BAFM, the IAFM utilizes DCN to learn the transformation offset of interlevel features, thereby achieving feature alignment and fusion between the low and high layers. In summary, the FA-BAFM is designed to capture the difference information between bitemporal features, while IAFM is designed to fuse interlevel features. They can learn different types of features from different views, thereby effectively enhancing the feature modeling capabilities of MAPNet.

D. Loss Function

In the prediction phase, the prediction head consists of a 3×3 convolution layer. Based on the architecture of the MAPNet, four predictions are generated and labeled as P_1, P_2, P_3, and P_4. Then, to calculate the loss function expediently, the different scales of prediction results are unified through an interpolation method. During the training phase, the weights of the model are optimized by minimizing the cross-entropy loss between the four prediction results and the groundtruth. The cross-entropy loss can be formulated as

$$L_{\mathrm{ce}}(Y, P) = -\frac{1}{N}\sum_{i=1}^{N}\left[Y_i \log_2 P_i + (1 - Y_i)\log_2(1 - P_i)\right] \tag{14}$$

where N is the total number of pixels in a change map, and P_i and Y_i are the prediction change map and corresponding

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the groundtruth, respectively. It should be noted that the above results serve as supervision for the MAPNet, providing auxiliary assistance to the model in capturing more effective features at multiple levels for subsequent predictions [10]. We employ deep supervision techniques for two main reasons. First, it helps mitigate the vanishing gradient problem, thereby enhancing the convergence of deep networks [20]. Second, by directly receiving feedback from the groundtruth, the hidden layers are guided to learn more discriminative feature representations [44], alleviating the burden on the decoder and further improving detection performance. The total loss can be formulated as

$$L_{\text{total}} = L_{\text{ce}}(\text{GT}, P_1) + L_{\text{ce}}(\text{GT}, P_2) + L_{\text{ce}}(\text{GT}, P_3) + L_{\text{ce}}(\text{GT}, P_4) \tag{15}$$

where P_1, P_2, P_3, and P_4 denote the probability maps in the final layers, and GT denotes the groundtruth. In the inference phase, only one prediction map P_1 is considered. An argmax operation is then applied to map the dual channel to a single channel, resulting in a binary matrix, indicating the corresponding pixel has changed or not.

IV. Experiments

A. Experimental Setting and Dataset Description

1) Experimental Setting: All models are implemented using PyTorch [45] with training and inference powered by a single NVIDIA GeForce RTX 2080Ti. In the training phase, the backbone of the MAPNet is initialized with the parameters of the mix transformer (MiT) [46] pretrained on ImageNet. For three datasets, consistent with other comparison experiments, the data enhancement, including randomly rotating, cropping, flipping, and distorting brightness, contrast, saturation, and hue, is performed during the training phase, and a patch size of 8×8 is utilized. The Adam [47] optimizer is employed to update network parameters with an initial learning rate set to 0.0001, weight decay set to 0.05, and parameters β_1 and β_2 set to 0.9 and 0.99, respectively. The training strategy combines linear and polynomial learning schemes, as illustrated in Fig. 6. A linear learning rate warm-up is used for the first 1000 iterations, followed by a learning scheduler from the 1000th to the 40 000th iteration with a period of 39 000 iterations. The deacy_lr is adjusted as

$$\text{deacy_lr} = \text{lr} \times \left(1.0 - \frac{\text{curent_iteration} - 1000}{\text{max_iteration}}\right)^{\text{power}} \tag{16}$$

where power and max_iteration are set to 1.0 and 40 000, respectively. curent_iteration represents the current iteration number, and deacy_lr is the learning rate for the current iteration. It is worth noting that during the training process, this study selects and saves the model with the highest F1-score, which is used as the final testing model.

2) Dataset Description:

a) LEVIR-CD [13] dataset: LEVIR-CD, a vast building CD dataset, originates on the Google Earth platform and comprises 637 pairs of HRRS images with a spatial resolution of 0.5 m/pixel; each image pair measures 1024×1024 pixels. These images capture diverse urban landscapes in 20 distinct regions in the state of Texas, USA, recording a total of 31 333 instances of architectural structures. The spectrum of architectural changes ranges from towering apartment complexes and villas to modest garages and expansive warehouses. Moreover, these variations extend beyond mere structural transitions, including subtle shifts induced by seasonal and lighting nuances. To better accommodate the requirements of model training and evaluation, and considering GPU memory constraints, we adhered to the official dataset split setting, which is 7:1:2. The original 1024×1024 pixel images are cropped into patches of size 256×256, with 7120, 1024, and 2048 patches allocated for training, validation, and testing, respectively. This cropping strategy is consistent with that adopted in [14].

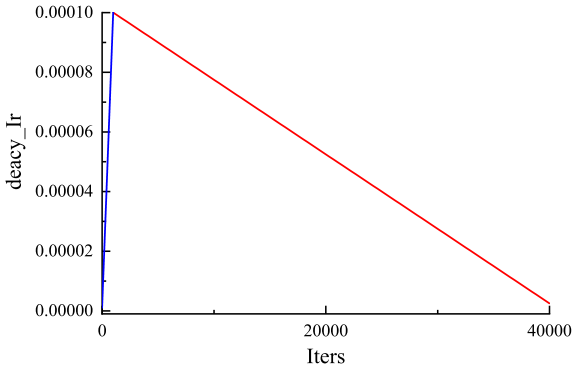


Fig. 6. deacy_lr curves of the network for each iteration. The blue line indicates the warm-up phase, while the red line indicates the normal phase.

b) The WHUBuilding CD dataset [48]: This dataset comprises a pair of high-resolution aerial images with dimensions of 32507×5354 pixels, boasting a spatial resolution of 0.3 m/pixel. These images chronicle the architectural changes in the Christchurch region of New Zealand following a 6.3-magnitude earthquake in 2011, capturing scenes from the years 2012 and 2016. The original images are cropped into 256×256 patch pairs with no overlap and then randomly divided into three subsets: 6096 pairs for training, 762 pairs for validation, and 762 pairs for testing. This splitting strategy aligns with that used in [10].

c) SYSU-CD dataset [49]: This dataset comprises 20 000 pairs of aerial images, each with spatial dimensions of 256×256 and a spatial resolution of 0.5 m. These images were captured over Hong Kong between 2007 and 2014, recording the rapid urbanization processes and diverse changes occurring during this period. It is noteworthy that the SYSU-CD dataset encompasses various types of targets, including newly constructed urban buildings, suburban expansion, preconstruction land preparation, vegetation changes, road expansion, and offshore construction, providing rich and diverse research material for CD. Distinguishing itself from the other two datasets, this dataset not only includes changes in the urban core but also records changes within suburban expansion and offshore construction, having more complex change scenarios. We adhere to the official splitting setting, allocating 8000, 4000, and 8000 samples for training, validation, and testing, respectively.

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TABLE I
ABLATION STUDY ON THE STRUCTURE OF THE MAPNET (%)

Name	Components				SYSU-CD				LEVIR-CD				Efficiency	
	LFEM	PAFM	IAFM	FA-BAFM	Pre	Rec	F1	IoU	Pre	Rec	F1	IoU	Params(M)	FLOPS(G)
Base	×	×	×	×	87.49	74.75	80.62	67.53	92.06	88.75	90.37	82.44	**3.92**	**1.52**
	√	×	×	×	**88.29**	74.61	80.88	67.89	92.37	89.31	90.81	83.17	4.11	2.21
	√	√	×	×	88.17	77.77	82.65	70.42	92.83	90.14	91.47	84.28	4.78	3.16
	√	√	√	×	88.26	78.05	82.88	70.70	**93.86**	89.36	91.55	84.42	4.87	3.17
	√	√	√	√	86.53	**81.03**	**83.69**	**71.95**	92.54	**91.27**	**91.90**	**85.01**	5.12	4.48

3) Evaluation Metrics: To assess the effectiveness of the proposed methods, five standard metrics are used: precision (Pre), recall (Rec), F1-score, intersection over union (IoU), and overall accuracy (OA). The Pre gauges the false alarm rate, while the Rec measures the missed alarm rate. The F1-score, as a harmonic mean of precision and recall, offers a comprehensive evaluation, with a higher score indicating a superior model [50]. Generally, OA assesses the overall correctness of predictions across all classes, while IoU measures the overlap between the change class on the detection map and the ground truth. These metrics are defined as follows:

$$\text{Pre} = \frac{\text{TP}}{\text{TP}+\text{FP}} \tag{17}$$

$$\text{Rec} = \frac{\text{TP}}{\text{TP}+\text{FN}} \tag{18}$$

$$\text{F1} = 2 \cdot \frac{\text{Pre} \cdot \text{Rec}}{\text{Pre}+\text{Rec}} \tag{19}$$

$$\text{OA} = \frac{\text{TP}+\text{TN}}{\text{TP}+\text{FP}+\text{TN}+\text{FN}} \tag{20}$$

$$\text{IoU} = \frac{\text{TP}}{\text{TP}+\text{FP}} \tag{21}$$

where TP represents the number of true positives, TN represents the number of true negatives, FP represents the number of false positives, and FN represents the number of false negatives.

B. Ablation Experiment

In this section, to assess the effectiveness of each proposed module, we conducted an ablation analysis of the core components of the proposed method. The "Base" represents a fundamental comparison network without the proposed modules in this article, and the "√" signifies the inclusion of corresponding modules. For example, the "Base" with the LFEM included is denoted by "√" under "LFEM," and other modules are similarly presented. First, as shown in Table I, with the addition of LFEM, all evaluation metrics exhibit an increasing trend. The IoU values of the model increased by 0.36% and 0.70% on the two datasets, respectively. The inclusion of LFEM helps the network capture both global and local features, enhancing its ability to identify changes. Second, when the PAFM is integrated, multiscale difference information is fused. This results in an increase in the F1-score by 1.77% and 0.66% on the two datasets, respectively. Then, with the inclusion of IAFM, the precision (Pre) reaches almost 94%, while the recall (Rec) decreases to 89.36% on the LEVIR-CD dataset, revealing that the IAFM is more sensitive to false alarms. Finally, the integration of FA-BAFM mitigates the effect of pseudo-changes and results in an improvement in the detection results. The experimental results show an increase of 0.81% and 0.35% in F1-score, and a corresponding rise of 1.91% and 2.98% in recall on the two datasets, respectively.

TABLE II
ABLATION STUDY ON THE FA-BAFM (%)

Model	LEVIR-CD		SYSU-CD	
	F1	IoU	F1	IoU
FDAF	91.70	84.68	81.83	69.25
FA-BAFM	**91.90**	**85.01**	**83.69**	**71.95**
w/o Attention	91.55	84.42	82.96	70.88
CBAM	91.75	84.76	83.37	71.48
CSAM	**91.90**	**85.01**	**83.69**	**71.95**

TABLE III
ABLATION STUDY ON THE PAFM (%)

Model	LEVIR-CD		SYSU-CD	
	F1	IoU	F1	IoU
w/o Max-pool	91.34	84.07	82.87	70.75
{1, 1, 1, 1}	91.63	84.55	82.72	70.52
{8, 4, 2, 1}	**91.90**	**85.01**	**83.69**	**71.95**

TABLE IV
ABLATION STUDY ON SUPERVISION (%)

Model	LEVIR-CD		SYSU-CD	
	F1	IoU	F1	IoU
P1	90.38	82.46	80.01	66.68
P1+P4	90.53	82.70	82.87	70.75
P1+P4+P3	91.50	84.34	82.92	70.82
P1+P4+P3+P2	**91.90**	**85.01**	**83.69**	**71.95**

After performing ablation experiments on the overall network structure, we further delve into the influence of key parameters and modules through detailed experiments. The outcomes of these experiments on the LEVIR-CD and SYSU-CD datasets are presented in Tables II–IV. The ablation experiments are conducted as follows.

1) The Ablation of the FA-BAFM: To assess the effectiveness of the difference feature modeling method in FA-BAFM, as shown in Table II, we replaced FA-BAFM with FDAF [51] which is a difference feature acquisition module based on feature alignment. From observations, it is evident that FA-BAFM captures more precise difference information and excels in F1-score and IoU, primarily attributed to its utilization of

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TABLE V
QUANTITATIVE COMPARISONS ON THREE CD DATASETS (%)

Model	LEVIR-CD					WHU					SYSU-CD				
	Pre	Rec	F1	IoU	OA	Pre	Rec	F1	IoU	OA	Pre	Rec	F1	IoU	OA
FC-EF	88.88	80.41	84.43	73.06	98.49	81.05	78.83	79.93	66.56	98.09	78.87	74.06	76.39	61.80	89.20
FC-Siam-Diff	88.47	83.44	85.88	75.26	98.60	64.56	86.44	73.91	58.62	97.06	67.73	75.48	71.39	55.51	85.74
FC-Siam-Conc	91.18	85.57	88.29	79.03	98.84	69.87	83.65	76.14	61.47	97.47	83.20	75.21	78.96	65.23	90.56
IFNet	90.11	90.59	90.35	82.40	99.01	92.77	91.09	91.92	85.05	99.21	85.59	76.14	80.59	67.49	91.35
BIT	91.45	89.03	90.22	82.19	99.01	89.35	86.64	87.85	78.33	98.84	78.99	75.46	77.18	62.84	89.49
SNUNet	92.53	90.08	91.29	83.97	99.12	82.91	90.18	86.39	76.04	98.63	84.03	75.16	79.34	65.76	90.77
Changeformer	92.23	89.16	90.67	82.93	99.07	95.95	90.43	93.11	87.11	99.35	**88.10**	75.10	81.08	68.19	91.74
HANet	92.29	88.27	90.24	82.21	99.03	92.49	88.86	90.64	82.88	99.12	82.95	75.86	79.25	65.63	90.63
DMINet	92.52	89.95	90.71	82.99	99.07	93.75	**92.96**	93.36	87.54	99.36	85.03	79.86	82.08	69.60	91.62
MAPNet	**92.54**	**91.27**	**91.90**	**85.01**	**99.18**	**96.60**	91.90	**94.19**	**89.02**	**99.45**	86.53	**81.03**	**83.69**	**71.95**	**92.55**

optical flow and attention-guided feature alignment. To further evaluate the effectiveness of attention mechanisms, the CSAM is removed, as shown in Table II. The absence of CSAM leads to a decrease in F1-score, Rec, and IoU, indicating that the attention mechanism contributes to the extraction of difference information, mainly thanks to its focus on the change areas. By replacing the CSAM with CBAM [52], a classic attention module, the outcomes demonstrate the superior performance of CSAM over CBAM when integrated into the FA-BAFM.

2) The Ablation of PDFFM: To evaluate the effectiveness of max-pooling and atrous rates in PAFM, we conduct ablation experiments, as depicted in Table III. First, by removing the max-pooling layer, there is a significant decrease in both F1-score and IoU, indicating its positive impact on PAFM. Subsequently, we experimented with different atrous rates within PAFM, including {8, 4, 2, 1} and {1, 1, 1, 1}. By adjusting these atrous rates, we found that the best performance is achieved with the original setting {8, 4, 2, 1}, demonstrating the effectiveness of progressively increasing the receptive field from large to small atrous rates.

3) The Ablation of Deep Supervision: To assess the effectiveness of deep supervision in network optimization, we conduct an ablation study on the extent of supervision. "P1," "P2," "P3," and "P4" represent the prediction results at different stages using deep supervision, as shown in Table IV. By adjusting the number of reference objects, the detection performance is boosted, mainly thanks to direct feedback from the groundtruth. This results in the hidden layers being regulated to learn more discriminative feature representations.

C. Comparison With State-of-the-Art Methods

A comparative analysis of the MAPNet is conducted against nine state-of-the-art (SOTA) CD approaches, including FC-EF [11], FC-Siam-diff [11], FC-Siam-conc [11], IFNNet [44], BIT [26], SNUNet [21], Changeformer [12], HANet [53], and DMINet [14]. Among these, FC-EF, FC-Siam-conc, and FC-Siam-diff are classic deep learning-based methods in the CD task. IFNet is a deeply supervised image fusion network that directly introduces change map losses into intermediate layers. SNUNet is a densely connected Siamese network designed to mitigate the loss of localization information. BIT and Changeformer are transformer-based approaches, while HANet and DMINet incorporate attention mechanisms into their models. Both of these methods have been recently proposed and have shown effectiveness by achieving competitive results on classic CD datasets. To maintain a fair comparison, we reimplement all these comparative methods using modified code based on their suggested parameters. For each method, we selected a set of optimal hyperparameters that maximized the F1-score on the validation subset.

1) Quantitative Evaluation: The overall comparison results of MAPNet with other methods on the LEVIR-CD, WHU, and SYSU-CD test sets are presented in Table V. The best values within each column are highlighted in bold, while the second-best values are underscored. The quantitative results demonstrate that MAPNet significantly outperforms other comparative methods on these datasets. Taking the LEVIR-CD dataset as an example, the proposed method achieves 92.54% in Pre, 91.27% in Rec, 91.90% in F1-score, 85.01% in IoU, and 99.18% in OA. Compared to CNN-based methods like FC-EF, FC-Siam-Diff, FC-Siam-Conc, and IFNet, the proposed method excels in all evaluation metrics, mainly thanks to its capability to extract both global and local information. In comparison to BIT and Changeformer, the proposed method shows an increase of 1.78% and 1.23% in F1-score, respectively, primarily due to the incorporation of multiscale and multilevel difference information. Finally, compared to SNUNet, HANet, and DMINet, the proposed method outperforms by 0.71%, 1.76%, and 1.29% in F1-score, respectively, primarily attributed to the effectiveness of feature alignment in mitigating pseudo-changes.

2) Qualitative Evaluation: The qualitative evaluation comparisons of different methods for three datasets are depicted in Figs. 7–9. In these figures, true negative (TN), false positive (FP), true positive (TP), and false negative (FN) regions are depicted in white, red, green, and black, respectively. Output change maps of FC-EF, FC-Siam-conc, and FC-Siam-diff are not provided because these methods yielded less competitive results in our experiments. From Figs. 7–9, it can be observed that the proposed method exhibits advantages in different aspects.

1) Merits of mitigating pseudo-changes: This is common in CD task, which can easily lead to false network detection. Take the LEVIR-CD (Fig. 7) as an example, where both

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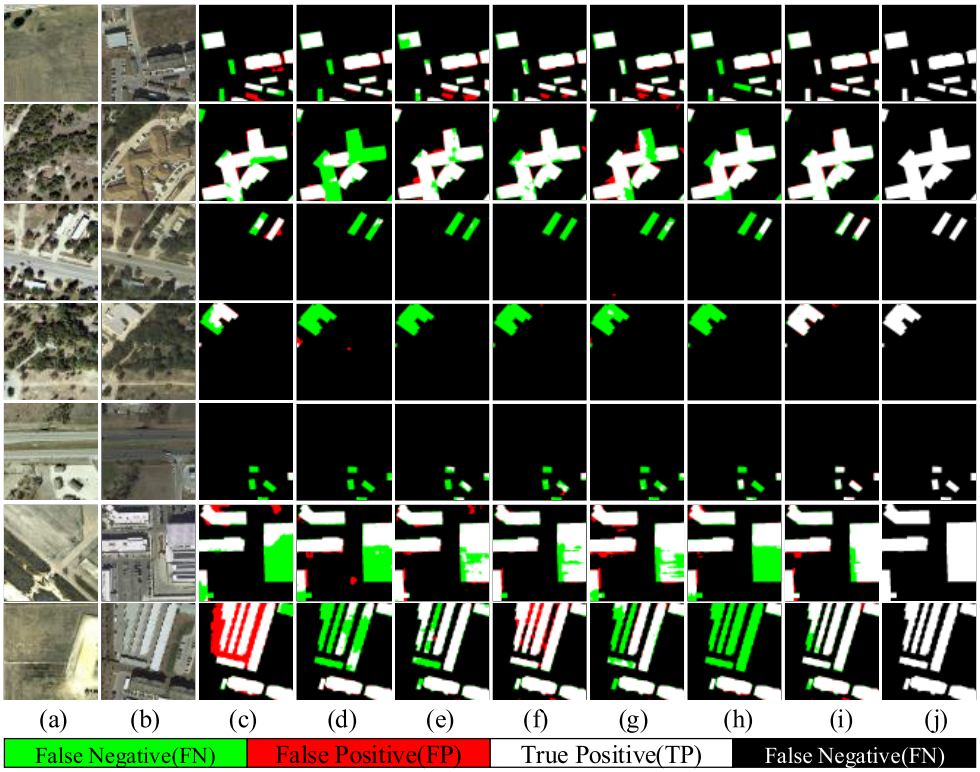


Fig. 7. Qualitative results of different methods on the LEVIR-CD dataset (a) RGB image of the first date, (b) RGB image of the second date, (c) IFNet, (d) BIT, (e) SNUNet, (f) Changeformer, (g) HANet, (h) DMINet, (i) MAPNet, and (j) groundtruth.

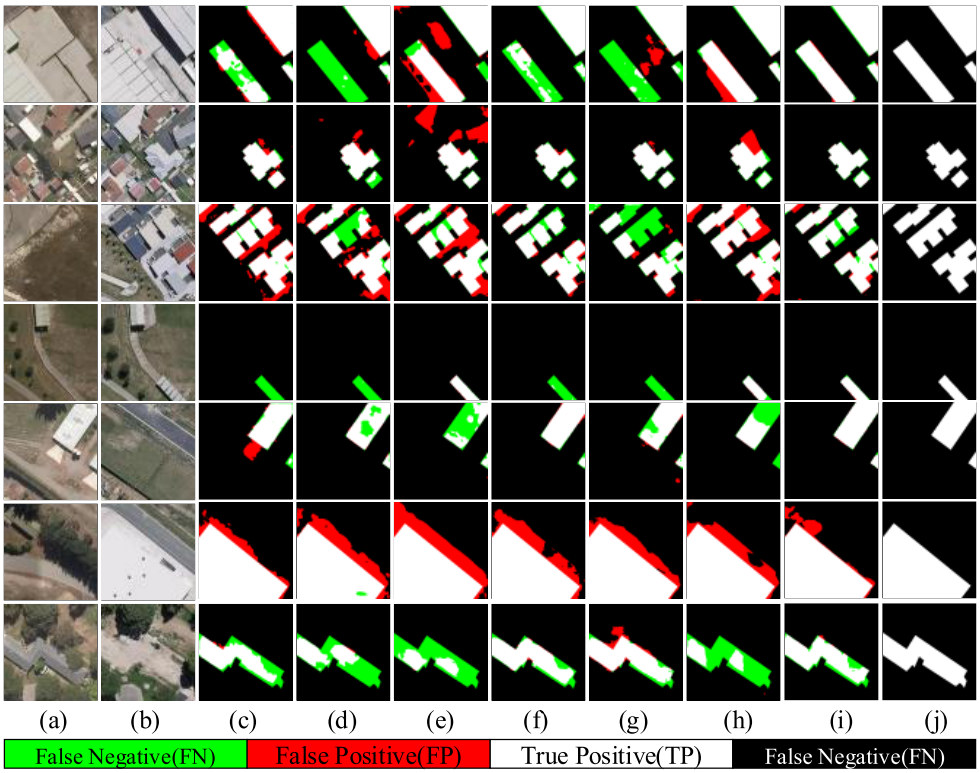


Fig. 8. Qualitative results of different methods on the WHU dataset (a) RGB image of the first date, (b) RGB image of the second date, (c) IFNet, (d) BIT, (e) SNUNet, (f) Changeformer, (g) HANet, (h) DMINet, (i) MAPNet, and (j) groundtruth.

IFNet and Changeformer are influenced by pseudo-changes, leading to incorrect detections. The interference worsens when changes and pseudo-change areas appear alternately, resulting in missed detections on BIT, SNUNet, HANet, and DMINet. The proposed method exhibits better visual results compared to other methods. Not only does the MAPNet accurately

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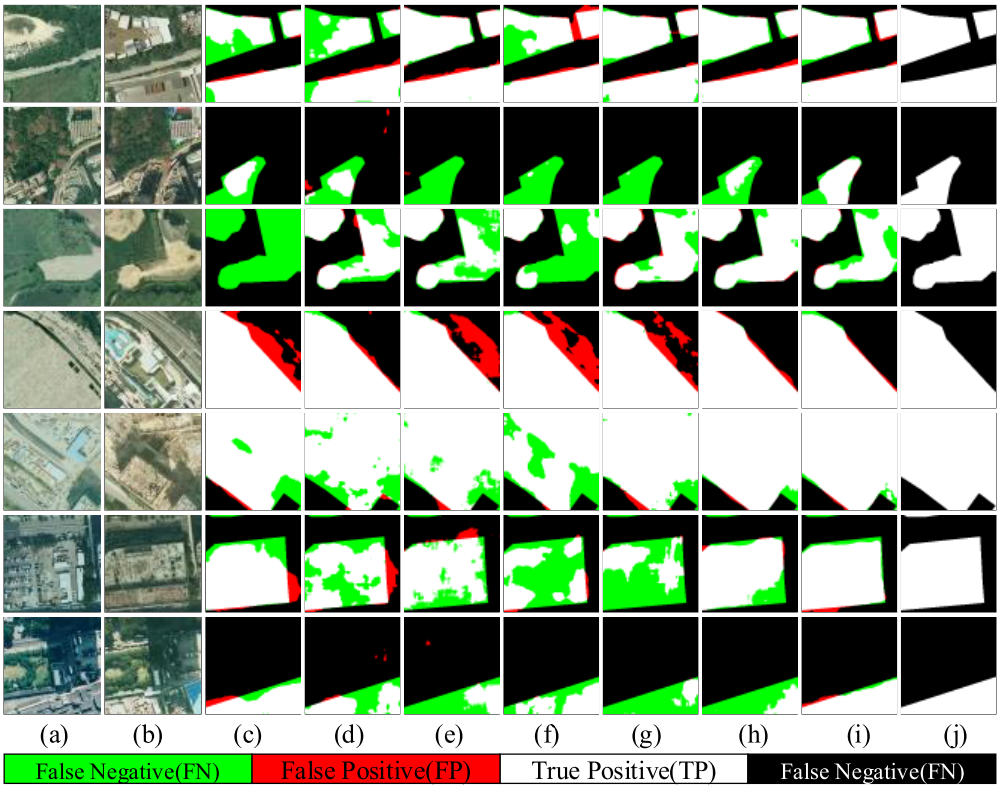


Fig. 9. Qualitative results of different methods on the SYSU-CD dataset (a) RGB image of the first date, (b) RGB image of the second date, (c) IFNet, (d) BIT, (e) SNUNet, (f) Changeformer, (g) HANet, (h) DMINet, (i) MAPNet, and (j) groundtruth.

delineate changing boundaries, but it also effectively alleviates pseudo-changes in the detection results. This is primarily attributed to the inclusion of FA-BAFM, which mitigates the impact of pseudo-changes and extracts more reliable difference information.

2) Merits of change location: Background interference is widespread and affects the ability to localize changing areas. Take the WHU (Fig. 8) as an example. Compared methods, such as IFNet, SNUNet, BIT, HANet, Changeformer, and DMINet, face challenges in accurately detecting the location of changes. Compared with other methods, the proposed method excels at locating changed objects and effectively suppressing background interference in bitemporal images. The main factor contributing to this improvement is the simultaneous capture of both global and local features, facilitating more accurate localization of change regions.

3) Merits of change integrity: The large-scope change areas are widespread among various datasets. Take the SYSU-CD (Fig. 9) as an example. Compared methods such as IFNet, BIT, SNUNet, Changeformer, and HANet may face challenges in capturing complex changes. However, the proposed method effectively addresses this issue by ensuring the integrity of the change area. This is primarily attributed to the PDFFM, which enables a detailed mapping of change information by effectively integrating difference features on different scales and levels.

D. Visualization on Feature Representation

In this section, to visualize the intermediate feature of the proposed method, the output probability maps at different stages are displayed, as illustrated in Fig. 10. Starting with a pair of bitemporal inputs, T1 and T2 [Fig. 10(a)], the probability maps after the backbone are shown in Fig. 10(b) and (c), labeled as B1 and B2, respectively. Second, the probability maps from the LFEM are shown in Fig. 10(d) and (e), labeled as L1 and L2, respectively. It is notable that, in comparison to B1 and B2, the position of changed targets appears clearer in L1 and L2, primarily due to the powerful local sensing capability of the LFEM. Additionally, the shallower layers retain finer spatial details, whereas the deeper layers depict essential semantic objects. Then, it can be observed that the difference probability maps from the FA-BAFM are represented as D1 [Fig. 10(f)]. Compared with L1 and L2, D1 records more information about the change and shows the location and contour of the change. This indicates that the network has already detected the changed targets, although the contours of these difference probability maps are not sufficiently clear. Subsequently, the probability maps from the PDFFM are depicted as D2 [Fig. 10(g)]. In contrast to D1, D2 exhibits clearer contours and a more complete change area, primarily due to the fusion of low-level detailed information and high-level semantic features. The final multilevel change probability maps, represented as P1 [Fig. 10(i)], demonstrate precise target localization and clear boundaries, which illustrate the effectiveness of the proposed method.

To showcase the efficacy of feature alignment in mitigating pseudo-changes, we visualize the prediction maps generated by the MAPNet. This process is achieved by adjusting the FA-BAFM, which is based on feature alignment, and the visualized results are illustrated in Fig. 11. The bitemporal

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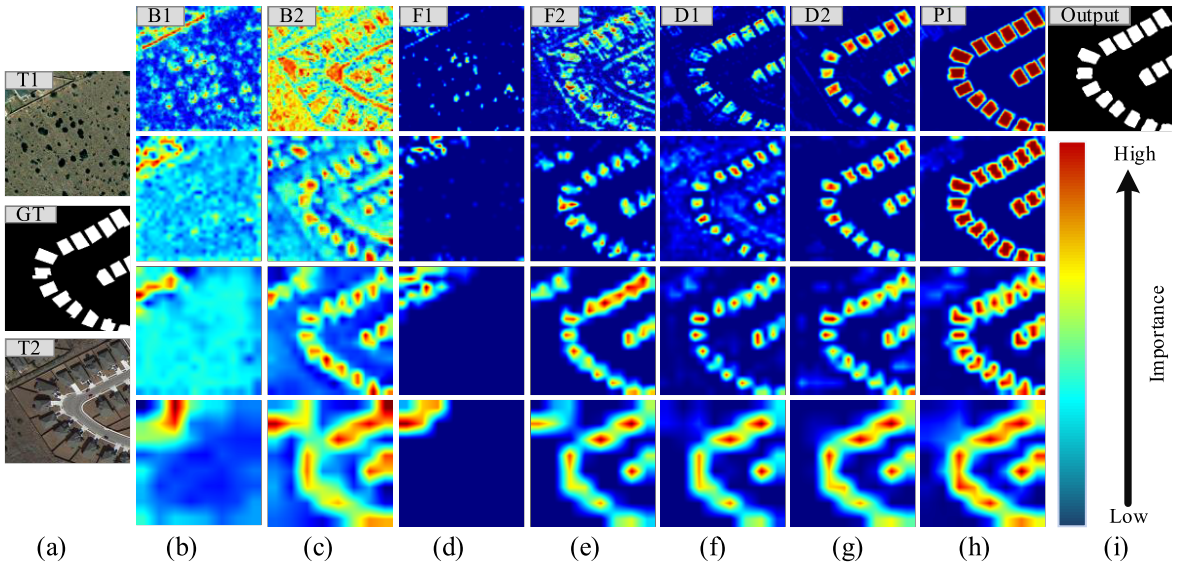


Fig. 10. Visualization results of the feature maps (a) RGB images of the first date, the second date, and the groundtruth, (b) output probability maps of the first date from the backbone, (c) output probability maps of the second date from backbone, (d) output probability maps of the first date from LFEM, (e) output probability maps of the second date from LFEM, (f) output probability maps of FA-BAFM, (g) output probability maps of PDFFM, (h) output probability maps of prediction, and (i) output of the MAPNet. The red color indicates a high pixel value in the visualization results, while the blue color indicates a low pixel value.

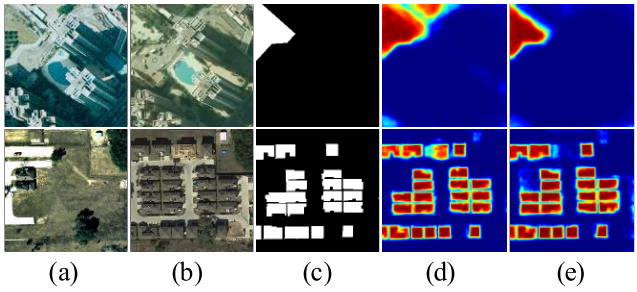


Fig. 11. Visualization results of features alignment. (a) RGB images of the first date, (b) RGB images of the second date, (c) ground truth, (d) probability maps of prediction without FA-BAFM, and (e) probability maps of prediction with FA-BAFM.

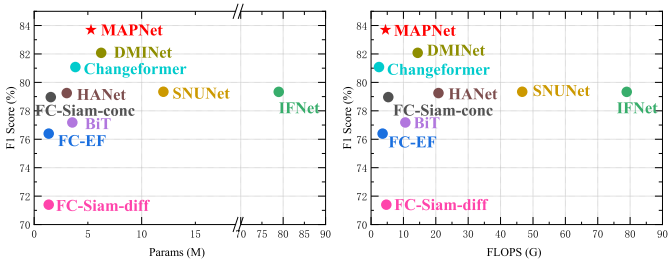


Fig. 12. F1-score versus Params and FLOPs on the SYSU-CD dataset.

images from the two datasets are depicted in Fig. 11(a) and (b), respectively. Compared with the groundtruth in Fig. 11(c), the first set of images exposes pseudo-changes induced by abnormal shadows resulting from alterations in lighting angles. In the second set, distinctions in the ground are incorrectly identified as changes in buildings. The prediction results of the MAPNet with FA-BAFM [Fig. 11(e)] closely resemble the groundtruth, affirming the contribution of feature alignment in effectively suppressing pseudo-changes.

E. Computational Complexity Evaluation

In this section, we conduct a benchmark of the MAPNet against nine SOTA CD methods. The floating-point operations per second (FLOPs) and parameters (Params) serve as widely used metrics for assessing computational cost. In this article, FLOPs and Params are reported from the inference on a256 $\times$ 256 image, and the F1-score is calculated on the test set of SYSU-CD. The scatter plot depicting F1-score versus Params and FLOPs is shown in Fig. 12. It reveals that DMINet achieves a commendable F1-score but demands a larger computational load. SNUNet, Changeformer, HANet, and BIT have moderate computational overhead but do not achieve top performance. IFNet has the highest number of calculations, but its accuracy is essentially flat compared to SNUNet and still has a gap with MAPNet. Although FC-Siam-conc, FC-Siam-diff, and FC-EF utilize minimal computational resources, their detection accuracy lags that of the other methods. In contrast, the proposed MAPNet outperforms all methods in terms of F1-score without burdening a significant computational burden.

V. Conclusion

In order to mitigate the disturbance of pseudo-changes and effectively model difference information, we propose the MAPNet for HRRS images CD in this article, with a focus on acquiring and integrating difference information. First, the designed LFEM extracts the global and local features from bitemporal images by combining the transformer and CNN. Second, different from the traditional difference information capture strategy, a flow and attention-guided bitemporal alignment fusion module is designed to reduce the impact of pseudo-changes in the process of obtaining difference

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information. Finally, the designed PDFFM progressively integrates multiscale and multilevel difference information, which is crucial for facing complex changing scenarios. We have conducted extensive experiments on three publicly available datasets, namely LEVIR-CD, WHU, and SYSU-CD, demonstrating that the MAPNet can achieve SOTA performance compared with other emerging competitors.

Notably, although the proposed method can alleviate the pseudo-changes problem effectively, there are still directions worth further research. First, multisource heterogeneous data will be considered for the CD task to address the problem of incomplete information about single-source data. Further, we consider introducing deep multiview learning to utilize the characteristics of multisource data. Finally, expert knowledge will be considered to assist data-driven neural networks, such as artificial experience, constraint rules, and regional information.

REFERENCES

[1] L. Bruzzone and F. Bovolo, "A novel framework for the design of change-detection systems for very-high-resolution remote sensing images," *Proc. IEEE*, vol. 101, no. 3, pp. 609–630, Mar. 2013.

[2] Z. Lv, M. Zhang, W. Sun, J. A. Benediktsson, T. Lei, and N. Falco, "Spatial-contextual information utilization framework for land cover change detection with hyperspectral remote sensed images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4411911.

[3] A. A. Abuelgasim et al., "Change detection using adaptive fuzzy neural networks: Environmental damage assessment after the Gulf War," *Remote Sens. Environ.*, vol. 70, no. 2, pp. 208–223, 1999.

[4] A. Frick and S. Tervooren, "A framework for the long-term monitoring of urban green volume based on multi-temporal and multi-sensoral remote sensing data," *J. Geovisualization Spatial Anal.*, vol. 3, no. 1, p. 6, Jun. 2019.

[5] P. Coppin, I. Jonckheere, K. Nackaerts, B. Muys, and E. Lambin, "Review ArticleDigital change detection methods in ecosystem monitoring: A review," *Int. J. Remote Sens.*, vol. 25, no. 9, pp. 1565–1596, May 2004.

[6] O. R. A. El-Kawy, J. K. Rød, H. A. Ismail, and A. S. Suliman, "Land use and land cover change detection in the western Nile delta of Egypt using remote sensing data," *Appl. Geography*, vol. 31, no. 2, pp. 483–494, Apr. 2011.

[7] D.-H. Kim et al., "Global, Landsat-based forest-cover change from 1990 to 2000," *Remote Sens. Environ.*, vol. 155, pp. 178–193, Dec. 2014.

[8] L. Khelifi and M. Mignotte, "Deep learning for change detection in remote sensing images: Comprehensive review and meta-analysis," *IEEE Access*, vol. 8, pp. 126385–126400, 2020.

[9] W. Shi, M. Zhang, R. Zhang, S. Chen, and Z. Zhan, "Change detection based on artificial intelligence: State-of-the-art and challenges," *Remote Sens.*, vol. 12, no. 10, p. 1688, May 2020.

[10] M. Lin, G. Yang, and H. Zhang, "Transition is a process: Pair-to-video change detection networks for very high resolution remote sensing images," *IEEE Trans. Image Process.*, vol. 32, pp. 57–71, 2023.

[11] R. C. Daudt, B. Le Saux, and A. Boulch, "Fully convolutional Siamese networks for change detection," in *Proc. 25th IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2018, pp. 4063–4067.

[12] W. G. C. Bandara and V. M. Patel, "A transformer-based Siamese network for change detection," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2022, pp. 207–210.

[13] H. Chen and Z. Shi, "A spatial–temporal attention-based method and a new dataset for remote sensing image change detection," *Remote Sens.*, vol. 12, no. 10, p. 1662, May 2020.

[14] Y. Feng, J. Jiang, H. Xu, and J. Zheng, "Change detection on remote sensing images using dual-branch multilevel intertemporal network," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4401015.

[15] X. Zhang, S. Cheng, L. Wang, and H. Li, "Asymmetric cross-attention hierarchical network based on CNN and transformer for bitemporal remote sensing images change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 2000415.

[16] V. Badrinarayanan, A. Kendall, and R. Cipolla, "SegNet: A deep convolutional encoder–decoder architecture for image segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 12, pp. 2481–2495, Dec. 2017.

[17] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. 18th Int. Med. Image Comput. Comput.-Assist. Intervent (MICCAI)*, 2015, pp. 234–241.

[18] R. C. Daudt, B. Le Saux, A. Boulch, and Y. Gousseau, "Multitask learning for large-scale semantic change detection," *Comput. Vis. Image Understand.*, vol. 187, Oct. 2019, Art. no. 102783.

[19] Z. Zhou, M. Siddiquee, N. Tajbakhsh, and J. Liang, "UNet++: A nested U-net architecture for medical image segmentation," in *Proc. Int. Workshop Deep Learn. Med. Image Anal.*, vol. 11045, 2018, pp. 3–11.

[20] D. Peng, Y. Zhang, and H. Guan, "End-to-end change detection for high resolution satellite images using improved UNet++," *Remote Sens.*, vol. 11, no. 11, p. 1382, Jun. 2019.

[21] S. Fang, K. Li, J. Shao, and Z. Li, "SNUNet-CD: A densely connected Siamese network for change detection of VHR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[22] H. Li, F. Zhu, X. Zheng, M. Liu, and G. Chen, "MSCDUNet: A deep learning framework for built-up area change detection integrating multispectral, SAR, and VHR data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 5163–5176, 2022.

[23] X. Wu, P. Gamba, J. Feng, R. Shang, X. Zhang, and L. Jiao, "A multitask framework for hyperspectral change detection and band reweighting with unbalanced contrastive learning," *IEEE Trans. Geosci. Remote Sens.*, early access, Jun. 5, 2024, doi: 10.1109/TGRS.2024.3410131.

[24] F. Luo, T. Zhou, J. Liu, T. Guo, X. Gong, and J. Ren, "Multiscale diff-changed feature fusion network for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5502713.

[25] Y. Teng, S. Liu, W. Sun, H. Yang, B. Wang, and J. Jia, "A VHR bi-temporal remote-sensing image change detection network based on Swin transformer," *Remote Sens.*, vol. 15, no. 10, p. 2645, May 2023.

[26] H. Chen, Z. Qi, and Z. Shi, "Remote sensing image change detection with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5607514.

[27] N. Shi, K. Chen, and G. Zhou, "A divided spatial and temporal context network for remote sensing change detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 4897–4908, 2022.

[28] C. Zhang, L. Wang, S. Cheng, and Y. Li, "SwinSUNet: Pure transformer network for remote sensing image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5224713.

[29] Z. Liu et al., "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 9992–10002.

[30] Y. Feng, H. Xu, J. Jiang, H. Liu, and J. Zheng, "ICIF-Net: Intra-scale cross-interaction and inter-scale feature fusion network for bitemporal remote sensing images change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4410213.

[31] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[32] W. Wang et al., "Pyramid vision transformer: A versatile backbone for dense prediction without convolutions," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 548–558.

[33] S. Ioffe and C. Szegedy, "Batch normalization: Accelerating deep network training by reducing internal covariate shift," in *Proc. Int. Conf. Mach. Learn.*, 2015, pp. 448–456.

[34] J. Dai et al., "Deformable convolutional networks," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 764–773.

[35] J. Lin, Y. Huang, and L. Wang, "FDAN: Flow-guided deformable alignment network for video super-resolution," 2021, *arXiv:2105.05640*.

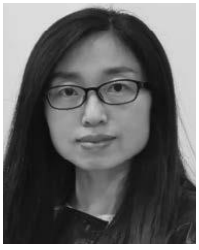
[36] S. Wang et al., "Align, perturb and decouple: Toward better leverage of difference information for RSI change detection," 2023, *arXiv:2305.18714*.

[37] Z. Huang, Y. Wei, X. Wang, W. Liu, T. S. Huang, and H. Shi, "AlignSeg: Feature-aligned segmentation networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 1, pp. 550–557, Jan. 2022.

[38] H. Zhou, F. Luo, H. Zhuang, Z. Weng, X. Gong, and Z. Lin, "Attention multihop graph and multiscale convolutional fusion network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5508614.

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[39] T. Guo, R. Wang, F. Luo, X. Gong, L. Zhang, and X. Gao, "Dual-view spectral and global spatial feature fusion network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5512913.
[40] L.-C. Chen, G. Papandreou, I. Kokkinos, K. Murphy, and A. L. Yuille, "DeepLab: Semantic image segmentation with deep convolutional nets, atrous convolution, and fully connected CRFs," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 40, no. 4, pp. 834–848, Apr. 2018.
[41] P. Wang et al., "Understanding convolution for semantic segmentation," in *Proc. IEEE Winter Conf. Appl. Comput. Vis. (WACV)*, Lake Tahoe, NV, USA, Mar. 2018, pp. 1451–1460.
[42] J. Li, Y. Wen, and L. He, "SCConv: Spatial and channel reconstruction convolution for feature redundancy," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 6153–6162.
[43] S. Huang, Z. Lu, R. Cheng, and C. He, "FaPN: Feature-aligned pyramid network for dense image prediction," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 844–853.
[44] C. Zhang et al., "A deeply supervised image fusion network for change detection in high resolution bi-temporal remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 166, pp. 183–200, Aug. 2020.
[45] A. Paszke, "Pytorch: An imperative style, high-performance deep learning library," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 32, 2019, pp. 1–11.
[46] E. Xie et al., "SegFormer: Simple and efficient design for semantic segmentation with transformers," in *Proc. Adv. Neural Inf. Process. Syst. (NIPS)*, vol. 34, Dec. 2021, pp. 12077–12090.
[47] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," 2014, *arXiv:1412.6980*.
[48] S. Ji, S. Wei, and M. Lu, "Fully convolutional networks for multisource building extraction from an open aerial and satellite imagery data set," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 574–586, Jan. 2019.
[49] Q. Shi, M. Liu, S. Li, X. Liu, F. Wang, and L. Zhang, "A deeply supervised attention metric-based network and an open aerial image dataset for remote sensing change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5604816.
[50] T. Lei et al., "Ultralightweight spatial–spectral feature cooperation network for change detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4402114.
[51] S. Fang, K. Li, and Z. Li, "Changer: Feature interaction is what you need for change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5610111.
[52] S. Woo, J. Park, and J.-Y. Lee, "CBAM: Convolutional block attention module," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Sep. 2018, pp. 3–19.
[53] C. Han, C. Wu, H. Guo, M. Hu, and H. Chen, "HANet: A hierarchical attention network for change detection with bitemporal very-high-resolution remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 3867–3878, 2023.



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